Jacob Mooradian-Plascencia

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EDUCATION

University of California, Riverside | Bachelor of Science in Computer Science | Riverside, CA | June 2024

WORK EXPERIENCE

Climformatics | San Francisco, CA

Climformatics leverages climate science expertise to provide accurate, customized future climate predictions to empower business decisions and risk mitigation.

Software Engineer | Aug 2024 – Present

- Launched BestDay2Marry, a SaaS turning climate forecasts into wedding-planning insights, adopted by 1,000+ users
  - Developed dual product flows: consumer portal (custom preferences, gifting flows, interactive maps, side-by-side date matching) and business dashboard (venue insights, weather visualizations, KPIs) using React, JavaScript, and Python
  - Engineered backend services with FastAPI, Firebase Auth, and BigQuery SQL (window functions/filters) to compute “best-day” windows by location; implemented secure data storage, CRUD operations, and automated email/ticketing
  - Integrated Stripe billing with trials, checkout, and webhook idempotency via FastAPI for reliable subscriptions
  - Owned cloud deployment with GCP (VMs, staging/prod, DNS migration, load balancing), and implemented CI/CD pipelines using GitHub Actions, Cloud Build, and Cloud Triggers, delivering a production-ready SaaS platform
- Built a dynamic geospatial tiling platform to replace static XYZ PNG tiles, cutting storage costs and delivering near-instant map performance
  - Converted terabytes of static PNG tiles to Cloud-Optimized GeoTIFFs with on-demand rescaling and colormaps, reducing file count and storage footprint by 98%
  - Deployed a FastAPI + TiTiler service on Cloud Run behind global load balancing and Cloud CDN, enabling low-latency tile rendering for 5+ client dashboards
  - Implemented caching and concurrency optimizations in Python, and automated deployments with GitHub Actions and Cloud Build, achieving near-static load times and supporting client integrations
- Redesigned climate dashboards (CALSEED, Xcel Energy) to make data explorable and decision-ready, improving usability and stakeholder adoption
  - Centered map-first layouts with standardized controls in React (TypeScript) + Tailwind CSS, reducing clutter and highlighting geospatial overlays
  - Shipped new features: CSV/PNG exports, shareable URLs, KPI cards, time-series graphing with legends, error bars, and severity levels for fire indices through custom React components
  - Built scalable services in Python/FastAPI, deployed on Cloud Run, to fetch forecasts and compute KPIs, returning lean responses to keep views fast across locations and time windows

Software Engineer Intern | Oct 2023 – Apr 2024

- Built a prototype weather analytics dashboard that powered investor demos and stakeholder working sessions
  - Developed a full-stack React + FastAPI MVP with custom charts for ML-derived forecast features, enabling exploratory analysis and decision making
  - Wrote SQL window functions and CTEs to rank ideal days at specific coordinates, exposed results via REST for reproducible analysis
  - Designed Figma wireframes to define web flow and set development roadmap

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL
Frameworks & Libraries: React, FastAPI, Node/Express, Tailwind
Cloud & Data: Google Cloud Platform (Cloud Run, Cloud Functions, Cloud Scheduler, Cloud CDN, Load Balancing, BigQuery, Dataproc, GCS), Firebase Authentication, Stripe, Google Analytics
Databases: BigQuery, Firestore, PostgreSQL
DevOps & QA: GitHub Actions, Cloud Build, Docker, CI/CD, Postman